WTF is Artificial Intelligence Really? | Yann LeCun x Nikhil Kamath | People by WTF Ep #4

[Speaker 2] (0:31 - 0:52)

I thought we could use today to figure out, A, what is AI? How did we get here? What lies next?

As an Indian 20 year old who wants to build a business in AI, a career in AI, what do we do?

[Speaker 1] (0:52 - 0:54)

Today? Today. Right now?

Yeah.

[Speaker 2] (1:04 - 1:23)

Hi Yann, good morning. Thank you for doing this. The very first thing we like to do is get to know you a bit more, how you came to be what you are today.

Could you tell us a little bit about where you were born, where you grew up leading up to today?

[Speaker 1] (1:23 - 3:11)

So I grew up near Paris in the suburbs. My dad was an engineer and I learned almost everything from him and always was interested in science and technology since I was a little kid and always saw myself as perhaps becoming an engineer. I had no idea how you became a scientist, but I became interested in this afterwards.

What's the difference between an engineer and a scientist? Well, it's very difficult to define and very often you have to be a little bit of both. But in science you try to understand the world, in engineering you try to create new things, and very often if you want to understand the world you need to create new things.

The progress of science very much is linked with progress in technology that allows to collect data. You know, the invention of the telescope allowed the discovery of planets, and the planets are rotating around the sun and things like this, right? The microscope will open the door to all kinds of things.

So technology enables science and for the problem that really has been my obsession for a long time is discovering the mysteries of, uncovering the mysteries of intelligence, and as an engineer I think the only way to do this is to build a machine that is intelligent, right? So there's both a scientific aspect of understanding intelligence, what it is, at a theoretical level and more practical side of things, and then of course the consequences of building intelligent machines could be really important for humanity.

[Speaker 2] (3:12 - 3:15)

And school in Paris, studying what?

[Speaker 1] (3:15 - 4:06)

So I studied electrical engineering, but as I progressed in my studies I became more and more interested in sort of more fundamental questions in mathematics, physics and AI. I did not study computer science. Of course, there is always computers involved when you study electrical engineering, even in the 1980s and late 70s actually when I started.

But I got to do a few independent projects with mathematics professors on the questions of AI and things like that, and I really got hooked into research. I was, you know, my favorite activity is to build new things, invent new things and then understand new things in a new way.

[Speaker 2] (4:07 - 4:15)

When somebody says Godfather of AI, the term, how does it make you feel? What do you think about it?

[Speaker 1] (4:15 - 5:22)

I mean, I don't particularly like this term, you know, I live in New Jersey, Godfather in New Jersey means you belong to the mafia. I mean, science is never a sort of individual pursuit. You make progress by the collision of ideas from multiple people and you do make hypotheses and then you try to show that your hypothesis is correct by demonstrating that the idea you have, the mental model of what should work is correct by demonstrating that it works or doing some theory and things like that.

And it's not an isolated activity. So there's always a lot of people who have contributed to progress. But then because of the nature of how the world works, we only remember just a few people.

I think a lot of the credit should go to a lot more people. It's just that we don't have a good memory for attributing credit to a lot of people.

[Speaker 2] (5:22 - 5:29)

So how does it feel to be a teacher today, Jan? When you were at NYU, are you the celebrity at NYU?

[Speaker 1] (5:31 - 6:01)

Let's say over the last several years, students come up to me at the end of the class. I want to take selfies. So there's a little bit of that.

I think if you are in the same room with someone, I think it's important to sort of make the session interactive because otherwise you can just watch a video. So that's what I try to do, really sort of engage with the students.

[Speaker 2] (6:02 - 6:11)

Do you suspect being a hero in academia, in research, is much like being a hero in sports? Or entrepreneurship? Or do you think it's harder?

[Speaker 1] (6:13 - 6:20)

Okay, there's something I'm happy about. The fact that there can be heroes, you know, from in science.

[Speaker 2] (6:21 - 6:26)

One can argue there was Newton and Einstein and all these people.

[Speaker 1] (6:26 - 7:39)

Well, Newton was not really kind of a public figure. I think. Einstein was.

Einstein certainly was. Yeah. And to some extent, you know, other, some other scientists also were sort of minor celebrities.

So, I mean, I think some of that comes from, you know, scientific production. But frankly, there's a lot of people who have made scientific contributions that are completely unknown and which I find it a little sad. But I think a lot of people who have become prominent in science and technology is not just because of the science they've produced, but also because of their public stance and out there.

You know, one thing that perhaps differentiates me from other scientists or a little quieter is that I'm very present on social networks and I give public talks and I have strong opinions about not just technical issues, but also policy issues to some extent. So that, I think, amplifies a little bit the popularity or unpopularity in certain circles. I'm seen like a complete idiot.

[Speaker 2] (7:40 - 7:53)

I've watched a lot of your interviews over the last fortnight, last month, in fact. If you were to state three problems with the world from Jan's lens, what would they be?

[Speaker 1] (7:57 - 9:00)

So as a scientist, you try to establish causal models of the world, right? So there are effects that we're seeing. And then the question is, what is it caused by?

And almost for almost every problem that we have, the cause is really a lack of knowledge or intelligence by humans. We're making mistakes. We're making mistakes because we're not smart enough to figure out we have a problem, because we're not smart enough to figure out solutions.

We're not smart enough to organize ourselves to find solutions, right? So things like, I mean, climate change is a huge issue, right? And, you know, there are, you know, political issues with that and questions of organizing the world, the governments, etc., but also potentially technological solutions to climate change. And I wish we were, you know, smarter so that we could find solutions faster.

[Speaker 2] (9:01 - 9:04)

So are you saying humans don't know why we do what we do, and that's the problem?

[Speaker 1] (9:05 - 10:10)

No, I think the mistakes we're making is because if we were a little smarter, if we had a better mental model of how the world works, and that's a central question in AI as well, I think we could solve our problems better. We would take decisions that are more rational. And the big issue I see in the world today is people who are not interested in finding the facts, are not interested in educating themselves, or maybe they are, but they don't have the means to do this, they don't have access to information and knowledge.

So I think the best thing we can do, and maybe that's why, you know, I became a professor, is to make people smarter. And to some extent, that's the best reason also to work on AI, because AI is going to amplify human intelligence, I mean, the overall intelligence of humanity, if you want. So I think that that's the key to solving a lot of problems that we have.

[Speaker 2] (10:11 - 11:13)

So just to preface this conversation, I'm an idiot. When it comes to anything around AI or technology, there isn't much that I know. And I've tried to learn over the past, over the very recent past.

And I have a lot of curiosity for it, but I don't know enough about it. A lot of the people watching us today are wannabe entrepreneurs, primarily based out of India. A lot of us have heard conjecture around AI, we've heard about the edge cases, both on the positive side and the negative side.

And I thought we could use today to figure out, for all of us, A, what is AI? How did we get here? And what likely next, if I were to break today down into three parts?

Should we start with what is AI?

[Speaker 1] (11:14 - 13:11)

Okay, that's a good question. What is intelligence even? So in the history of AI, I think the problem of what is AI feels a little bit like the story of the blind man with the elephant, right?

That there are very different aspects to intelligence. And over the history of AI, people have addressed one view of what intelligence is, and basically ignored all the other aspects. So one of the early aspects of intelligence that people addressed with AI in the 1950s was, you know, intelligence is about reasoning.

How do we reason? So how do we reason logically? How do we search for solutions to a new problem?

And in the 50s, people figured out, well, we have a problem, let's say, that's become a starter problem in AI, or computer science now. I give you a bunch of cities, and I ask you, you have to go through every single city, and what's the shortest path, the shortest circuit to go around the city? That's called a traveling salesman problem.

And they say, like, every reasoning problem can be formulated in terms of searching for a solution to a problem. There's a space of possible solution. There is something that tells you whether you found a good solution or not, or some number that tells you the length of the path, and you just have to search for the shortest path, right?

And to some extent, you could reduce every reasoning problem to a problem of this type. In mathematics, we call this optimization. So you have a problem, you can evaluate whether your problem is solved or not with a number that indicates, you know, it's low if the length of your path is small, and it's high if it's longer.

And you search for a solution that minimizes that length of the path.

[Speaker 2] (13:11 - 13:20)

So is finding solutions related to intelligence? If you were to ask me, what is intelligence, I would be like, dumbfounded in trying to define it in a sentence.

[Speaker 1] (13:21 - 13:24)

Right. I mean, so that comes back to the elephant analogy.

[Speaker 2] (13:25 - 13:26)

Can you explain the elephant analogy?

[Speaker 1] (13:27 - 16:26)

Well, so you know, the blind man and the elephant, right? So the first blind man goes to the side of the elephant and says, that sounds like, looks like a wall. And then one goes to a leg that looks

like a tree, and the other one, you know, touches the trunk.

That's a, that's a pipe. And nobody has a complete picture of what an elephant is, right? And you see it from the various angles.

So this aspect of intelligence as being a search for a solution to a particular problem is, you know, a small piece of the elephant. It's one aspect of intelligence, but it's not, it's not the entire thing. But in the 50s, one branch of AI was basically only concerned by this.

And that branch was essentially dominant until, until the 1990s. That AI consists in searching for a solution, for plans, you know, if you want to, you know, stack a bunch of objects on top of each other, and some objects are bigger than others, you have to sort of organize the order in which you're going to stack the objects, you know, you search for a sequence of actions to arrive at a goal that's called planning. Or even, let's say, you have a robot arm, and you have to grab an object, but there is, there's, you know, obstacles in front of it, you have to plan a trajectory for the arm to grab, to grab the object.

So all of that is planning, that's part of this searching for a solution to a problem. But that part of AI, which again, was started in the 50s and was dominant until the 90s, completely ignored things like perception, like how do we understand the world? How do we recognize an object?

How do we separate an object from its background so we can identify it? And you know, how do we think, not in terms of logic or search, but perhaps in more abstract terms? So that was essentially ignored.

But there was another branch of AI, also started in the 50s, that said, well, let's try to reproduce the mechanisms of intelligence that we see in animals and humans. And animals and humans have brains. The brains basically organize themselves, they learn, right, they're not spontaneously smart.

And the intelligence is sort of an emergent phenomenon of networks of very simple elements in large numbers that are connected with each other. So in the 50s, 40s, people started discovering that intelligence and memory comes from the strength of the connections between neurons in a sort of simplified manner. And the way the brain learns is by modifying the strength of the connections between neurons.

So some people came up with sort of theoretical models and actually electronic circuits that reproduce this. You know, can we build...

[Speaker 2] (16:26 - 16:30)

So intelligence, you're saying, was largely the ability to solve a certain problem.

[Speaker 1] (16:30 - 16:41)

So that's the first view, right, to solve particular problems that were given. The second one is the ability to learn. Right.

Okay. And that created those two branches of AI.

[Speaker 2] (16:41 - 16:41)

Right.

[Speaker 1] (16:42 - 17:11)

So the one that started with the ability to learn, there was some success in the late 50s, early 60s, and it died in the late 60s because the type of learning procedures for those neural networks that people devised in the 60s turned out to be extremely limited. There was no way you could use this to produce truly intelligent machines. But it had a lot of consequences in various parts of engineering, a field of engineering called pattern recognition.

[Speaker 2] (17:12 - 17:17)

So you're saying now that intelligence is the ability of a system to learn as well?

[Speaker 1] (17:17 - 18:35)

To learn. And the simplest situation in which you need machines to learn is for perception, interpreting images, interpreting sounds. And what did computers use to do that?

So for that, it's basically what caused the emergence of what we could call classical computer science. You write a program and that program basically internally searches for a solution and has some way of checking whether the solution it proposes is good or not. People had a name for this in the 60s, they called this heuristic programming, because you can never exhaustively search all solutions for a good one because the number of solutions is ridiculously large.

You know, at chess, for example, you can play a certain number of moves, but then for every move that you play, your opponent can play a certain number of moves. And then for every of those moves, you can play a certain number of moves. So you get this exponential explosion of the number of possible trajectories, basically, or sequences of moves.

And you cannot possibly explore all of them until the end of the game to figure out which move to play first. So you have to use what was called heuristics to basically not search the entire graph or tree of possibilities.

[Speaker 2] (18:35 - 18:56)

So we'll put up a graph explaining this, but what you're saying in heuristics AI is you would have a user who would put in an input, there would be a bunch of rules, and you would use like a tree

search or an expert AI which would run a function like if this, then that, if not, then this, to try and get to an end state.

[Speaker 1] (18:56 - 20:10)

Yeah, so something that would be defined and the program would be completely written by a person. And the difference between a good and a bad system would be in how smart the system is in searching for a good solution without doing exhaustive search. OK, that's the heuristics part of it.

A slightly different approach is the one that's based on logic, right? So you have rules and facts. What other facts can you deduce from the existing fact and the rules, which would be logical formula and things like this?

That was, you know, pretty dominant in the 1980s, and that led to an area of AI called expert systems or world-based systems that to some extent is very connected with this idea of search. OK, and then in parallel to this, there is the bottom-up approach. You know, let's try to reproduce the, to some extent, get inspiration from the basic mechanisms of intelligence in biology, implement, allow machines to learn and basically organize themselves with the idea that...

[Speaker 2] (20:10 - 20:11)

How would you do that?

[Speaker 1] (20:12 - 23:14)

So it's based on the same idea that neuroscientists figured out was going on in the brain, which is that the learning mechanism in the brain proceeds by modification of the strength of the connections between neurons. Right. And people had imagined that this type of learning could actually be reproduced in machines.

So first there was the idea that you could, you know, that neurons were simple, simple computational elements. There were proposals around those lines in the 1940s by mathematicians like McCulloch and Pitts and people like that. And then in the 50s and early 60s, people proposed a very simple algorithm to change the strength of the connections between neurons so that they could learn the task.

So the first machine of this type was called a perceptron and it was proposed in 1957. It's a very simple thing and it's very simple to understand. Let's say you want to train a system to recognize simple shapes, images.

OK, what is an image for a computer or for an artificial system? It's an array of numbers. We know that today because we're familiar with digital cameras and pixels.

Right. So let's take a black and white camera. A pixel is, if the pixel is black, it's a zero.

If it's white, it's a one. OK, so it can take only two values, black or white. OK, if you want to build this with 1950s technology, you would put an array of photo sensors, photo cells, right, with the lens in front of them and you would show an image, very low resolution, maybe 20 by 20 pixels or something like this or even lower.

So now that gives you an array of numbers that you can feed to a computer. But what they did in the 1950s, computers were incredibly expensive. So they actually built electronic circuits.

So the pixels were voltages coming out of the photo sensors. And then you want to train a system to recognize simple shapes. Let's say distinguish the shape of a C from the shape of a D drawn on this on this array.

So you show an example of a C. And then you let the system produce an output. This output will also be a voltage.

And the way the output is going to be computed is a weighted sum of the of the values that come in of the pixels that are one or zero. The weights are connections to a simulated neuron, which is just an electronic circuit that computes, you know, if it's a one or a zero, we're going to multiply this one or the zero by a weight, which is like a resistor that you can change the value of. OK, and then all of the pixels with their weight are going to be summed up.

If the weighted sum is larger than the threshold, it's a C. If it's lower than that threshold, it's a D.

[Speaker 2] (23:15 - 23:17)

What era was this? Which year did you say?

[Speaker 1] (23:19 - 24:54)

1957. So now how do you train this? So training consists in changing the value of those weights.

You can have positive or negative weights. And what you do is you show a C and the system computes the weighted sum. So for a C, you want the weighted sum to be large, larger than zero, let's say.

OK, and let's say it's smaller than zero. So the system made a mistake. So you tell it, no, it should be larger.

OK, you press a button basically and you tell it, I really want the output to be bigger. So what the system does is that it changes all the weights that get a one. So it increases them a little bit.

If you increase all the weights that get a one, the weighted sum increases, right? And if you keep doing this, changing the weights just a little bit every time, eventually the weighted sum is going

to go above zero. And then the system will recognize this as a C.

And what did we use this for back in the 50s and 60s? So nothing really very practical other than recognizing simple shapes. So you repeat showing a C and a D.

And for the C, you say increase the weighted sum. For the D, you say decrease the weighted sum. So decrease the weights that have a one, increase the weights that have a zero.

And then eventually the system settles on the configuration of weights so that when you show a C, it's above the threshold. When you show a D, it's below the threshold so it can distinguish the two. And what it's going to do is, you know, give a positive weight to the pixels that only appear for the C and a negative weight to the pixels that only appear for the D.

And that will sort of discriminate between those two.

[Speaker 2] (24:54 - 25:01)

So we had heuristics AI, expert AI, trying to mimic biology, all of this in the 50s and 60s?

[Speaker 1] (25:01 - 25:22)

In the 50s, yeah, starting in the 50s. And then, you know, two different branches basically competing with each other. And they tried to kind of, so one person, a prominent figure in AI in the pioneering days is Marvin Minsky.

He was a professor at MIT.

[Speaker 2] (25:22 - 25:30)

He was a Marvin, there is a, I remember reading about this, there's a Marvin Clause or debate or something like that, right?

[Speaker 1] (25:31 - 26:29)

Well, he was, he had pretty strong opinions about things. So there was a lot of discussions. And he's interesting because he started his PhD in the 50s, trying to build neural nets, and then completely changed his mind and became basically a big advocate for the other approach, the more logic-based and search approach.

And in the late 60s or mid-60s, he wrote a book, co-wrote a book with Seymour Papert, who was a mathematician at MIT, whose title was Perceptron. And the whole book was to do some theory about Perceptron and to show that the capabilities of the Perceptron was limited. So the people who were working on neural net at the time kept working on neural net, but they changed the name of what they were doing.

They called it statistical pattern recognition, which sounds much more serious, or adaptive filter theory, which also sounds very serious. And those had enormous applications in the real world.

[Speaker 2] (26:30 - 26:48)

In my world, it's always been, I work in finance, and hedge funds and fund managers have always been attempting to pump a lot of data into a neural network to recognize patterns. Right. Is it the same thing that we're talking about, an evolution from the 50s?

[Speaker 1] (26:49 - 27:05)

Yeah, absolutely. I mean, the process I described of changing coefficients, you know, up or down to get the output you want, you could think of this as a iterative process very similar to linear regression, which, if you work in finance, you probably know that.

[Speaker 2] (27:06 - 27:25)

But what I've realized, Jan, is even today, it's very easy to tweak data that you have collected retrospectively to make something appear like it makes sense. But financial activity tends to be so random that I don't know if you can build a model based on that.

[Speaker 1] (27:26 - 28:32)

Right, so that addresses a bigger issue. When you train a system this way, right, so the generic principle, which is called supervised learning, is you give an input to the system, it produces an output. If the output is not the one you want, you adjust the coefficients so that the output gets closer to the one you want, okay?

And there are efficient ways to figure out how to tweak the parameters so that the output gets closer to the one you want. And if you keep doing this on hundreds, thousands, millions, billions of examples, eventually the system, if it's powerful enough, will figure it out. Now, the problem with the perceptron is that the type of functions, input-output functions, that was accessible to a perceptron was very limited.

So there was no way you could take a natural image, you know, a photo, and train the system to tell you if there is a dog or a cat or a table in it. That was just not possible. The system was not powerful enough to really compute this kind of complex function.

This is what neural nets and deep learning changed in the 1980s.

[Speaker 2] (28:34 - 28:48)

Just before you get into neural nets, if I'm trying to paint the entirety of the picture, would you say there is intelligence on top, artificial intelligence, and below that is machine learning and neural

nets are a part of machine learning?

[Speaker 1] (28:48 - 30:08)

Yeah, so in terms of fields and subfields, AI is more of a problem than a solution. It's a field of investigation. And then there is different techniques you can use for that, right?

So there is something that jokingly is referred to as good old-fashioned AI, Go-Fi, which is using logic and search and heuristic programming and things like this, which is what you will find in sort of standard textbooks on AI. Then there is machine learning. So there the idea is you don't completely program a machine to do something, you train it from data.

That means you need data. Within this there is a subcategory called deep learning, and this is the reason why we hear so much about AI in the last dozen years, is because of deep learning. And neural nets is really the ancestor of deep learning.

Deep learning is a new name for it, if you want. And then there is application areas. Below that.

Below that. And they can use combinations of those techniques, right? So big applications are computer vision, interpreting images, speech recognition, natural language understanding, and maybe speech synthesis also can be viewed as part of this, although it's more connected with signal processing.

And then various other applications. So time series prediction or financial modeling and things like this could be seen as part of this.

[Speaker 2] (30:08 - 30:22)

So I'm breaking it down. AI has Gophi under it, which is traditional in nature, like you explained. Then machine learning.

Can you define Gophi in a simple one-line definition?

[Speaker 1] (30:23 - 30:40)

So Gophi is the descendant of what I was describing earlier as searching for solutions, right? This idea that it's all about reasoning. Reasoning is all about search, looking for a solution to a problem and having a way of characterizing whether you've found a solution.

[Speaker 2] (30:40 - 30:47)

So you mean the rule-based thing, the input and an output, based on what rule applies like that?

[Speaker 1] (30:49 - 31:06)

Yeah, I mean any rule-based system, anything that uses logical inference, deducing facts from

rules and previous facts, searching for a solution, like finding the shortest path in a graph or something. Those are good old-fashioned AI.

[Speaker 2] (31:06 - 31:09)

And under machine learning, what are the different types of ML?

[Speaker 1] (31:10 - 32:06)

So there is so-called traditional machine learning. I'm not sure that deserves the term. And this is basically derived from statistical estimation.

So things like linear regression would be part of it. And then there are other methods that are slightly more sophisticated, boosting classification trees, super vector machines, kernel methods. I mean, there's a bunch of methods of this type in Bayesian inference that are part of machine learning in the sense that they obey that model of, you know, you build a program but the program is really not finished.

It's got a bunch of tunable parameters and the input-output function is determined by the value of those parameters. And so you train the system from data using this iterative adjustment techniques that I described before. Show examples.

If the answer is incorrect, adjust the parameters so that it comes closer to the answer you want.

[Speaker 2] (32:07 - 32:09)

So machine learning is supervised in a way?

[Speaker 1] (32:09 - 32:36)

So that's supervised learning, okay? You tell the system, here is an output, here is the desired output. But there are other forms of learning.

So one different form is reinforcement learning. So in reinforcement learning, you don't tell the system the correct answer. You just tell it whether the answer it produced was good or bad.

You give it a single number that tells it your answer was good or was bad.

[Speaker 2] (32:37 - 32:44)

And what happens next? Say I'm a reinforcement learning engine and you tell me an answer was good or bad. What do I do next?

[Speaker 1] (32:44 - 33:17)

Well, so if your answer was good, you don't do much. If your answer was bad, then you have to

figure out which answer, among all the possible answers I could have produced, which one would be a better one. So maybe you try another answer and you say, what about this one?

Is it better or is it worse? If the environment tells you it was better, then you kind of de-emphasize the first one and emphasize that one by tuning the parameters inside of a neural net or something like that, some sort of learning machine.

[Speaker 2] (33:18 - 33:20)

So what is self-supervised learning?

[Speaker 1] (33:20 - 33:37)

Okay, so self-supervised learning is what has become very prominent over the last five, six years and is really the main component or the main contribution to the success of things like chatbot and natural language understanding systems.

[Speaker 2] (33:37 - 33:39)

They don't fall under reinforcement learning?

[Speaker 1] (33:39 - 35:12)

No, it's more similar to supervised learning, but the difference is that instead of having a clear input and output and training the system to produce the output from the input, you basically only have things that can either be input and output. Let me take an example. You take a piece of text and you corrupt that text in some way by removing some words, right?

So now you have a partially masked text where some words are missing. And you train a machine to predict the words that are missing. So the technique you would use for this is supervised learning because you tell the system, here is the correct word that you should predict at that location.

And the system can use all the words that it can see to predict the words that it cannot see. And this is an example for supervised learning? Self-supervised learning.

It's self-supervised because there is no differentiation between input and output. It's really kind of the same thing. And if the input is, for example, an image, the way you would train a self-supervised learning system is that you would corrupt or transform the image in some way.

And then you would train the system to recover the original image from the corrupted or transformed version of it. Okay, so there is no supervision. You don't need someone to go through a few million images and labeling them, is it a cat or a dog or a table or a chair?

It's a task of basically understanding the input, the internal structure of the input, by being able to

fill in the blanks.

[Speaker 2] (35:12 - 35:30)

Forgive me for asking maybe a really stupid question. I'm trying to picture this. Let's say I have X amount of data.

I have 10 lines that say cats are black, dogs are white, whatever, 10 lines. I remove a part of it and then I tell the model to fill it in.

[Speaker 1] (35:30 - 35:30)

Yeah.

[Speaker 2] (35:31 - 35:36)

Are you saying at that point of time I also tell the model the answer, saying this should be the answer?

[Speaker 1] (35:36 - 35:43)

Yeah. You tell it, here is the answer that I removed. Can you predict this missing?

[Speaker 2] (35:43 - 35:48)

Can you arrive at the answer which I removed and I'm telling you that this was the answer?

[Speaker 1] (35:48 - 42:38)

Right. But you can only use the thing that you can see. So you don't see the answer on the input.

You have to predict it. But I'm telling you, during training, I tell you what it is. And so the system can adjust its parameters in a supervised fashion.

So the difference is not in the algorithms themselves. It's basically supervised learning, but it's in the structure of the system and the way the data is used and produced. You don't need to basically have someone going through millions of images and telling you this is a cat or a dog at a table.

You just show an image of a dog, a cat, or a table and you corrupt it, partially change it, change the colors maybe or something, and then ask the system to recover the original one from the corrupted one. So that's one particular form of self-supervised learning. And this is what's been incredibly successful for natural language understanding.

So chatbots or LLMs, large language models, are a special case of that, where you train a system to predict a word, but you only allow it to look at the words that precede it, that are to the

left of it. And that requires building the neural net in a particular way so that the connections that predict one word only look at the words that precede. So then you don't need to corrupt the input.

You just show an input, and through the structure of the system, the system is trained to predict the next word from the content. And these are all examples of neural networks in a way. These are all, underlying this, are a particular way of connecting neurons with each other, simulated neurons, right?

Or simple elements that compute a very simple mathematical function. Something like a weighted sum, and what's adjustable are the weights. Or in the case of transformer architectures, which are very popular at the moment, they consist in basically comparing every input to each other and producing weights.

I could explain this. It's a little more complicated. What is a transformer?

Okay, so there are several architectural components from which you can build a neural net. So let me start with a very simple idea. Let's say you want to build a neural net that recognizes images, okay?

So again, an image is an array of numbers indicating the brightness of every pixel, right? You can build a neural network with a single layer. So let's say you want to distinguish 10 categories, okay?

Cats, dogs, tables, and chairs, and cars, and whatever. Or let's say it's simpler. You want to recognize the 10 digits, okay?

0 through 9. Someone drawing a digit. It's drawn on a 16 by 16 pixel area, so you have 256 inputs, and you have 10 outputs.

Okay, you can have what's called a single layer neural net, which basically each output is a weighted sum of the pixels, and you try to train those weights in such a way that when you show a 0, the output 0 is the most active and the other ones are less active, and so forth for all the categories. Okay. That may work for simple shapes like printed digits.

It won't work for handwriting because there's so much variability in the characters that you cannot reduce the classification to a simple weighted sum. Okay, so the breakthrough that occurred in the 1980s was to stack multiple layers of neurons. So each neuron computes a weighted sum and then passes this weighted sum through essentially a threshold function.

So if the weighted sum is below a threshold, the neuron stays inactive, the output is 0. If it's above a threshold, it's active. Okay, there's various ways to do this.

But it's nonlinear, and that's very important. So you stack two layers, where the middle layer you could think of as detecting sort of basic motifs on the inputs, and then the second layer sort of

integrates those motifs to figure out, okay, this is a C because it's got two endpoints, you know. The shape of the C kind of stops there, and I can detect that.

So if there's two of them, that's a C. And the D doesn't, but the C has two corners. Maybe I can detect that.

The system spontaneously learns to do this from end to end. And the way it works is through an algorithm called backpropagation. And what this backpropagation algorithm does is that when you show an image of a C and you tell the system, this is a C, so activate this output neuron.

Do not activate the other ones. It knows how to adjust the parameters so that the output gets closer to the one you want. And that's done by propagating signals backwards to basically figure out the sensitivity of each output to each weight so that you can change the weights in such a way that the good output increases and the bad outputs decrease, right?

So that's backpropagation. That algorithm has to do this. So the backpropagation algorithm popped up in the 1980s conceptually.

It existed before, but people didn't realize they could use it for machine learning. And there was a wave of interest in neural nets starting in the mid-'80s, lasting 10, 15 years, to exploit this idea of multilayer networks. And this was crucial because it lifted some of the limitations that Minsky and Papert in the 60s said were, you know, the perceptron was subjected to.

So a big wave of interest. But then people realized that to train those neural nets, you need a lot of data. And this was before the internet.

There was not much data. You need fast computers, and computers were not that fast. So people kind of lost interest a little bit in this.

But one thing that I worked on in the late-'80s, early-'90s is if you want a system of this type to recognize images, you kind of have to connect the neurons to each other in a particular way that facilitates the system sort of paying attention, you know, being able to detect motifs, for example, right? Local motifs. So I got inspiration from biology again, the classical work in neuroscience that went back to the 1960s, to basically organize the way the neurons are connected to each other into layers so that they're biased towards kind of finding good solutions for image recognition.

So that's called a convolutional neural network, or CONNECT.

[Speaker 2] (42:40 - 43:01)

So just to come back to this, where you are, like, so you broke down machine learning. I'm sorry, I keep going back, or I'll get confused. So under machine learning, the really popular pathway right now, let's say self-supervised, which has GPD and a bunch of other things, what's happening in the reinforcement learning space?

[Speaker 1] (43:02 - 43:23)

So not so much anymore. There was a big wave of interest in reinforcement learning about a dozen years ago, and companies like DeepMind set themselves up with the idea that reinforcement learning was going to be the key element towards building truly intelligent machines.

[Speaker 2] (43:24 - 43:27)

Can you again define reinforcement learning once more in a line?

[Speaker 1] (43:27 - 44:19)

So reinforcement learning is a situation where you don't tell the system what the correct answer is. You just tell it whether the answer you produced was good or bad. So there are many possible answers.

It's very inefficient because the system has to try many things before it gets the correct answer. And so it's very inefficient. It requires many, many, many trials.

And so it works really well for games. It's very efficient. If you want to train a system to play chess or go or things like that, poker, reinforcement learning is great because you can have the system play millions of games against itself or copies of itself.

And it can adjust it. It wins or loses a game, so it knows which policy, which flavor of the neural net won the game and sort of reinforces that and de-emphasizes the one that lost. And so the system basically can train itself.

[Speaker 2] (44:19 - 44:21)

And what did you say a transformer was?

[Speaker 1] (44:22 - 48:06)

Okay, so I was coming to this through the convolutional net. So there is this particular way of connecting simulated neurons with each other to bias it towards doing a good job for certain types of data. And convolutional nets are really good for data that comes from the natural world, whether it's an image or an audio signal, which are things that are nearby values in the array of numbers that come to you in an image or audio signal.

Nearby values are generally very similar to each other. So if you take a picture, any picture, natural image, and you take two neighboring pixels, they're very likely to have the same color or the same intensity. Now what I'm talking about here is the fact that natural data, like images and audio and just about any natural signal, has some natural underlying structure to it.

And if you build a neural net in a particular way that can take advantage of this structure, it will learn faster. It will learn with fewer samples. So we started doing experiments with this in the late 80s and built those convolutional nets.

They were inspired by the architecture of the visual cortex, really. And there is some mathematical justification for it. But the basic idea is that each neuron in a convolutional net only looks at a small area of the image.

And you have multiple neurons looking at multiple areas of the image, and they all do the same thing. They all have the same weights. It's a basic concept which connects with the mathematical concept of convolutions, and so that's why those things are called convolutional nets.

Okay, so that's what's called an architectural component, or a module, a convolution. It's something that has an interesting property, which is that if you show it an input, it's going to produce a particular output. If you shift the input, the output will be shifted but otherwise unchanged.

And that's a very interesting property for audio signals, images, and various other natural signals. Okay, now a transformer is a different way of arranging the neurons, if you will, in such a way that the inputs are a number of different items. We call them tokens.

They're really vectors, which means list of numbers, okay? And the property of the layer or the block of a transformer is that if you permute the inputs, the output will be permuted similarly but otherwise unchanged. When you say otherwise unchanged, you mean?

What I mean is that if you give a bunch of tokens, you run through the transformer, you will get a bunch of output tokens, the same number generally as the number of input tokens. There will be different vectors. If you now take the first half and the second half of your sequence of input tokens and you flip them, what you will get is the same result that you got previously, but it will be flipped exactly the same way, okay?

So the input-output function is technically, we call this equivariant to permutation. So it basically views the inputs as a set in which the order of the object does not matter, okay? Convolutional nets, on the other hand, view the input as something where an object could appear at any location on the input and it shouldn't make any difference to the output, or the output should change but otherwise, I mean, should shift but otherwise stay unchanged.

That's equivariance to translation. Now, when you build a neural net, you basically combine components of this type so that you get the property you want out of the entire neural net. So you combine things like convolutions and transformer blocks.

[Speaker 2] (48:06 - 48:10)

What is a convolution, Jan? I'm sorry. I'm going to ask you to simplify every single term.

[Speaker 1] (48:10 - 49:06)

Oh, absolutely. So a convolution is this component for a convolutional neural net. So the idea of it is that you have a neuron that looks at a part of the input, and then you have another neuron that looks at another part of the input, but it computes the same function as the first neuron.

And then you replicate that same neuron for every location on the input so that you can think of each of those neurons as detecting a particular motif on a part of the input and all the neurons detecting the same motif at different parts of the input. So that now if you take an input and you shift it, you're going to get the same output shifted because you're going to have the same neurons detecting the same motif just at different locations. So that's what gives you this shift equivariance.

That's a convolution. Mathematically, there's something called a convolution that mathematicians invented a long time ago, and that's basically what this does.

[Speaker 2] (49:07 - 49:13)

When you say neuron in all of this, can you explain the basis of just that term? What is it?

[Speaker 1] (49:13 - 50:11)

So we use that term. It's an abuse of language because those neurons are not really neurons like in the brain. They are to real neurons as an airplane wing is to a bird wing.

So it has the same concept. And what a neuron does in a neural net is computing a weighted sum of its inputs and then comparing that weighted sum to a threshold, activating the output if it's above the threshold and producing zero if it's below the threshold. That's the basic neuron.

Now, there are variations of this. In a transformer, it's a slightly different type of mathematics. You're kind of comparing vectors to each other and things like that.

But that's kind of the basic functionality of a neuron. It's a combination of a linear operation where you have coefficients that you can change the value of through training, and then a nonlinear function, a threshold, or something like that, that detects something or not.

[Speaker 2] (50:11 - 50:22)

Right. We looked online, and while we were researching, we could not find a good definition for neural network language model and how it works in simple terms.

[Speaker 1] (50:23 - 51:36)

Okay. So the idea of a language model goes back to the 1940s. A gentleman called Claude

Shannon, he's a very famous mathematician who used to work at Bel Air, where I used to work, although he wasn't there anymore when I joined.

And he came up with a theory called information theory and then was fascinated by the idea that you could discover the structure in data. So he invented something where you take a text and you say, I'm giving you a sequence of letters, and I'm asking you, what is the next letter that comes afterwards? Okay.

So let's take an English word or whatever in a sort of, let's say, Roman language. If you have a series of letters and the last one is a Q, it's very likely that the next letter is a U. You almost never have a Q without a U behind it, unless it's an Arabic word or something that's been transliterated.

So for every letter that you observe, you can build a table of the probabilities that the next letter will be an A, a B, a C.

[Speaker 2] (51:37 - 51:39)

This is where the word generative comes from?

[Speaker 1] (51:40 - 54:08)

Yeah, so it's generative because if you have this table of what we call conditional probabilities, right, given the previous letter, what is the probability of the next letter? You can use this to generate text. You start with a letter, let's say Q, okay, and then you look through the table of probability.

What's the next letter that is most likely? You just pick that one. That's going to be U.

Or you don't pick that one. You pick the next letter with the probability that is, you flip a coin, or you generate a random number on your computer, and then you produce the following letter according to the probabilities that you measured on real text. And you keep doing this.

And the system is going to just generate letters. It's not going to look like words. It's probably not even going to be pronounceable.

But if instead of a context of one letter, you take a context of two letters, then it becomes kind of more readable. It's still not words, right? If you take a context of three letters, then it becomes even nicer.

And as you increase the size of the context that determines the probability of the next letter, it becomes more and more readable. But you have an issue there, which is that the size of the table you need, if you look at the first letter and you have to figure out what's the probability for the next letter, you need a table of 26 rows and 26 columns. For each first letter, what is the probability for every possible second letter?

So it's a table of 26 by 26. Now, if the context has two letters, now the number of rows in your table is 26 squared, because you have 26 squared possible sequences of two letters, right? And for each of those, you need 26 probabilities.

So it's 26 cubed, the size of your table. As you add characters, the table increases to 26 to the power n, where n is the length of the sequence. So that's called an n-gram model, and that's a language model.

You can do this at the level of characters. It's more difficult to do this at the level of a word, because you might have 100,000 possible words, right? So now your table is gigantic.

So you can train a word model or language model by just filling up this table of probabilities by training on a large corpus of text. But it becomes impractical above a certain length of context.

[Speaker 2] (54:08 - 54:11)

Because of the amount of compute and work required.

[Speaker 1] (54:11 - 54:30)

Also, it's the memory of storing all those tables, and also the fact that those tables are going to be very sparsely populated, because you can have billions of words of text. Most combinations of words don't appear. Some of them are extremely rare, and so you cannot estimate the probability properly, okay?

[Speaker 2] (54:32 - 54:34)

Is this a part of self-supervised learning?

[Speaker 1] (54:34 - 57:48)

So you could think of this as an instance of self-supervised learning, because you only need sequences of symbols and it doesn't matter where they come from. They don't necessarily come from human production if they are not text. It could be, for example, a sequence of frames for a video, right?

I mean, you would have to turn it into discrete objects, which of course is difficult, but it's whatever data comes to you. So in the late 90s, some people had the idea that, in particular Yoshua Bengio, the idea that you could use a neural net to do this prediction. Instead of filling up tables with conditional probabilities that you measure from text, just train a neural net to predict the next word, okay?

Give it a context of words and just train it to produce a probability distribution over the next word. And he experimented with this with, you know, neural nets that were big for the time but small by

today's standards. And one difficulty was you cannot exactly predict what word is going to come next, so you have to produce a probability over all the words.

And there's maybe 100,000 words in a typical language. And so that means you need to output 100,000 scores, one for each word that indicates with which probability that word follows the previous sequence of words. So he demonstrated that that could work.

And, you know, even with the computers of the time, it was kind of challenging, but it could work. And then the idea was kind of revived more recently. And it turned out that if you use those transformer architectures, which I didn't explain, and you train them on basically the entirety of all the publicly available text on the internet, and you build the architecture of those systems so that it's trained to, you know, take a context of words and predict the next word, and if you make the context potentially very large, something like a few thousand, a few tens of thousands, or even a million words, then you get systems that seem to have emergent property, that they can answer questions. They can, you know, if you make them really big, they have so many parameters that are adjustable, they may have tens of billions or hundreds of billions of parameters.

That gives them a large amount of memory, and they seem able to store a lot of knowledge about the data they've been trained on. If it's text, it will regurgitate solutions to puzzles. They will, you know, give you answers to questions you may have.

It's mostly retrieval. There's a very tiny bit of reasoning, but really not much, and that's an important limitation. But it's still surprising how well those things work, and it's, you know, what people got really surprised about is that those systems can manipulate language in ways that are very impressive, right?

I mean, humans have pretty limited in the way we manipulate language, and those things seem to be really good at it. I mean, they capture grammar and, you know, syntax and everything in multiple languages, right? That's pretty amazing.

[Speaker 2] (57:48 - 58:20)

So if I were to, like, go back and paint a tree, so let's say AI on top, machine learning under it. I'm talking about what is making the news today and what everybody's so excited about. Machine learning has different things, different neural networks under it.

There's a reinforcement one like DeepMind. There is a self-supervised generative chat GPT because using it as a placeholder, as it's the most popular one right now.

[Speaker 1] (58:21 - 59:55)

LLM. Autoregressive LLM, really, that's what it should be called. Autoregressive LLM.

Yeah, I mean, the proper organization is, yeah, there is AI at the top. Machine learning is a

particular way of approaching the AI problem. Under this is deep learning, which is really the foundation of pretty much all of AI today.

So basically neural networks with multiple layers, right? So the idea of this goes back to the 1980s and back propagation. That's still the basic foundation of everything we do.

Under this there are several families of architectures, convolutional nets, transformers, combinations thereof. Then there is, under transformers, there is several flavors of it, some of which can be applied to image recognition or audio, some of which can be applied to representing natural language, but not generating it. And then there's a subcategory, natural language models, which are autoregressive transformers.

So transformers have a particular architecture that allows them to predict the next word, and then you can use it to just generate word, because given a sequence of words, it's been trained to produce the next word. So given a text, you have to produce the next word, and then you shift the input by one. So now the word you generated is part of its input, and you can ask it to generate the second word.

Shift that, third word. Shift that, fourth word. That's autoregressive prediction.

It's the same concept as autoregressive models in finance and econometrics and stuff like that. Same stuff.

[Speaker 2] (59:56 - 1:00:02)

And these work best for text, but not for pictures, videos, or any of that?

[Speaker 1] (1:00:02 - 1:01:38)

That's right. And the reason it works for text and not for other things is because text is discrete, so there is a finite number of possible things that can happen, right? There's a finite number of words in the dictionary.

So if you can discretize your signal, then you can use those autoregressive prediction systems. And the main issue is that you're never going to be able to make an exact prediction, and so the system is going to have to learn some sort of probability distribution, or at least produce scores that are different for different potential outputs. So you can output a list of probabilities if you have a finite number of possibilities, which is the case for language.

But if you want to predict what is going to happen in a video, the number of possible frames, video frames, is essentially infinite, right? You have, you know, let's say a million pixels, right, but an image is a thousand pixels. Pixels are in color, so you have three values.

So that's three million values that you have to produce. And we don't know how to represent a

probability distribution over the set of all possible images with three million pixels. But this is what everybody is very excited about.

This is what a lot of us consider the next challenge in AI. So basically you have systems that can learn how the world works by watching videos.

[Speaker 2] (1:01:38 - 1:01:49)

And if you were to say videos learn from videos and pictures, which will be the next phase, where does that fall in this entire equation? Does it come under where LLMs sit today?

[Speaker 1] (1:01:49 - 1:03:10)

No, it's completely different from LLM, which is why I've been pretty vocal about the fact that LLMs are not the path to human-level intelligence. LLMs work for discrete worlds. They don't work for continuous high-dimensional worlds, which is the case for video.

And this is why LLMs do not understand the physical world and cannot be used in their current form to really understand the physical world. So we have, I mean, LLMs are amazing in their ability to manipulate language, but they can make very, very stupid mistakes that reveal they really don't understand how the world works, the underlying world. And this is why we have systems that can pass the bar exam or write an essay for you.

But we don't have domestic robots. We don't have self-driving cars or completely autonomous level five self-driving cars. We don't have systems that really understand very basic things that your cat can understand.

So I've been kind of vocal in saying that the smartest LLMs are not as smart as your house cat. And it's really true. So the challenge for the next few years is to build AI systems that lift the limitations of LLMs. So systems that understand the physical world have persistent memory, which LLMs really don't have at the moment.

[Speaker 2] (1:03:11 - 1:03:12)

Persistent memory.

[Speaker 1] (1:03:12 - 1:03:20)

Persistent memory, which means they can remember things, right? Store facts in a memory and then retrieve them when it's interesting.

[Speaker 2] (1:03:21 - 1:03:23)

Can't LLMs remember stuff now?

[Speaker 1] (1:03:23 - 1:04:45)

The only memory that an LLM— there's two types of memory that an LLM has. The first type is in the parameters, in the coefficients that are adjusted during training, right? So they will learn something.

It's not really kind of storing a piece of information. If you train an LLM on a bunch of novels, it cannot regurgitate the novels. But it will remember something about the statistics of the words in that novel, and it might be able to answer questions, general questions about the story and things like this, but it's not going to be able to regurgitate all the words, right?

Kind of like humans, right? You read a novel, you can't remember all the words unless you spend a lot of efforts trying to do this. So that's the first type of memory.

And then the second memory is the context, the prompt that you type. And since the system can generate word and those words or those tokens are injected in its input, it can use this as some sort of working memory, but it's a very limited form of memory. What you want is a memory that would be more similar to what we have in our brains, what mammalians have, called a hippocampus.

A hippocampus is a kind of brain structure in the center of the brain, inner bits of the cortex. And if you don't have a hippocampus, you can't remember things for more than about 90 seconds.

[Speaker 2] (1:04:45 - 1:05:01)

And if you were to draw a path from intelligence that we described on top, all the way down to self-supervised learning, how do you suspect that path will look towards us getting to the point where we are learning from videos and images and more human-like intelligence?

[Speaker 1] (1:05:01 - 1:05:42)

So the path that I've been trying to plot is discovering new architectures, different from those autoregressive architectures used for LLM, that would be applicable to video, so that self-supervised learning could be used to train those systems. And this type of self-supervised learning basically would be, here is a piece of a video, predict what comes next. And if a system can do a good job at predicting what's going to happen next in a video, that means it probably has understood a lot about the underlying structure of the world.

Similarly to a large language model learns a lot about a language by just being trained to predict the next word.

[Speaker 2] (1:05:43 - 1:05:47)

Not like I will understand, but if you had to give us a line on how that architecture might look.

[Speaker 1] (1:05:47 - 1:08:24)

Okay, so here is the issue. Because, as I told you, those autoregressive architectures work for text because text is discrete, and you can never predict what comes next. We can produce a probability distribution over what comes next.

You cannot do this for images and video because it's just too complicated mathematically, and you can show that it's intractable and blah, blah, blah. So predicting all the pixels in a video that follow a particular video segment basically is not possible. Or not possible to a degree that would be useful for the problem that we're interested in.

What we want is a system that has the ability to predict what's going to happen in the world because that's a good way for a system to be able to plan. If I can plan that, if I push my hand to this glass and I close my hand and I lift it up, I'm going to grab the glass and I can drink. I can plan a sequence of actions to arrive at a particular result.

So I have a good model of the world that says the state of the world at time t is this. The glass is on the table. The action I'm going to take is close my hand around it, and lift.

What is going to be the state of the world at time t plus three seconds after I close my hand and lifted my arm? And the state of the world is going to be I'm going to have that glass in my hand. So if you have this kind of world model, state of the world, action, next state of the world, then you can imagine, you can predict, you can predict the outcome of a sequence of actions.

You can imagine taking a sequence of actions and then predicting in your mind what the outcome will be. You can predict if this outcome is something that satisfies a goal that you want to accomplish, like drink a little bit of water, take a sip. And what you can do is through search, so now we are connecting with old AI, search a sequence of actions that will actually satisfy this goal.

So this is the type of reasoning and planning that psychologists call System 2. Daniel Kahneman is a late Nobel Prize winning psychologist and he makes this distinction between System 1 and System 2, where System 1 is actions you can take without thinking, subconscious, it's just reactive. And the System 2 is what you have to deliberately plan and think about to be able to produce an action or a sequence of actions.

[Speaker 2] (1:08:24 - 1:08:30)

So Jan, will memory eventually be the answer? Because as humans from biology, we learn through memory.

[Speaker 1] (1:08:31 - 1:10:33)

Well, it depends what type of memory. I mean, we also have multiple types of memory. We have

the hippocampus that I mentioned.

So hippocampus is used to store long-term memories, like, you know, things that happened to you when you were a child and things like that. Basic facts about the world, like, you know, when your mom was born or something. You know, also, you know, which way you came in here, so where is the door?

So this is more recent, short-term memory. Episodic memory, working memory, so if you're thinking about something, you're kind of manipulating things in your head, you have to kind of temporarily store data. That's the hippocampus.

And your cortex does the computation and basically reads from this memory and updates it. It's very much like a bit of a computer where the cortex is the CPU and the hippocampus is the memory that you read from and write into. But the current design of AI systems is not like that.

So LLMs do not have a separate memory other than the prompt that you can generate token in. And they don't have this ability to search through a set of answers for which one is the correct one, although they're starting to have that to some extent. So you may have heard of O1 from OpenAI.

There is kind of similar work at Meta and other places where there's sort of very basic forms of reasoning that consist in having an LLM produce lots of different sequences of words and then having a way of searching through this word which one is the best. But it's very inefficient. So ultimately that's not what you want.

So going back to the question of how do we get machines to learn by observing the world, learning from video, we cannot use the architectures that are generative, that just produce every pixel in the video. That's just completely impractical. And I've tried to do this for almost 15 years.

And five years ago we came up with a different way of doing things that are called JEPA. So it's a different architecture, and that means Joint Embedding Predictive Architecture.

[Speaker 2] (1:10:34 - 1:10:40)

And what it means is— I watched this for a long time on your Lex Friedman interview when you spoke about JEPA and I still don't get it.

[Speaker 1] (1:10:41 - 1:11:16)

Okay, here's a basic idea. Tell me if you don't understand because I can explain it in different ways. Instead of taking a piece of video and training a big neural net to predict all the pixels of the continuation of that video, you take the video and you run it through an encoder, which is going to be a big neural net that's going to produce an abstract representation of the video.

And then you take the reminder of the video, the future, the second half of that video, run it through the same encoder. And then you train a prediction system which is similar, perhaps, to an LLM.

[Speaker 2] (1:11:16 - 1:11:19)

Much like LLM where you delete a part of the data to train the model?

[Speaker 1] (1:11:19 - 1:11:56)

That's right. So an LLM, you take a piece of text and you train it to predict the reminder of the text. And you do this word by word, but you could predict multiple words.

So here we're going to do the same thing. We're going to take a video and then train a system to predict the reminder of the video. But instead of predicting all the pixels in the video, we're going to run those videos through encoders which are going to compute abstract representations of the video.

And we're going to do the prediction in that space of representations. So instead of predicting pixels, we predict abstract representations of those pixels where all the things that are basically unpredictable have been eliminated from the representation.

[Speaker 2] (1:11:57 - 1:12:08)

Is that a bit like also predicting tomorrow? Because if I were to video my life up until now and run it through the encoder, it will give me some kind of representation of tomorrow?

[Speaker 1] (1:12:08 - 1:12:32)

Well, yes, but at an abstract level, right? So you can predict. You're based in Bangalore, I heard.

So at some point you're going to fly back to Bangalore. And you can predict how long it's going to take to go back to Bangalore, but you cannot predict all the details of what will happen during your journey back to Bangalore, exactly how long it's going to take, given traffic.

[Speaker 2] (1:12:32 - 1:12:38)

How far can you extrapolate? What will happen three months from now if I have video data of the last 10 years of my life?

[Speaker 1] (1:12:39 - 1:12:52)

So here's a trick. It's an interesting question. You can predict very long term, but the longer in the future you can predict, the more abstract the representation level at which you can make the prediction.

[Speaker 2] (1:12:52 - 1:13:12)

Let me ask you a question. If you were to extrapolate 50 years forward, all of our lives, you figure out how to build this architecture and it's implemented and it's working, where video of our life up until now has been programmed into it and we're trying to predict 50 years forward, what do you suspect you will see? Climate change and world war?

[Speaker 1] (1:13:13 - 1:14:12)

So what I see is, okay, so there is a plan for the next few years to build systems that can understand the world from video. Perhaps what they'll be able to learn are those world models, which are action conditions, so they will be able to imagine what the consequence of an action or a sequence of action will be. They'll be perhaps able to plan complex sequences of action hierarchically because those world models will be hierarchical.

They will have world models that can predict really short term, make accurate prediction, but only in the short term. Like if I move my muscle in this particular way, my arm is going to be in this particular location about a millisecond from now. That's really short range, but very precise.

And then longer time prediction, if I go to the airport, catch a plane, I'll be in Paris tomorrow morning. Or if I study and I get good grades in college, I can have a good life or something. So you can make long term prediction and design plans that would satisfy certain criteria that you have.

[Speaker 2] (1:14:13 - 1:14:17)

If AI were to predict the future, would it be utopian or dystopian?

[Speaker 1] (1:14:17 - 1:15:21)

It would be utopian, because it would be just an alternative way for predicting the future than our brains and for planning action sequences to satisfy certain conditions to achieve goals. That is, alternative to using our brains, perhaps accumulating more knowledge to be able to do this, and perhaps having abilities that humans don't have because of the limitations of our brain, right? Computers can calculate and stuff like that, right?

So the future is that if we succeed in this plan, which may succeed within the next five or ten years, five to ten years, we'll have systems that, as time goes by, we can build up to become as intelligent as humans, perhaps. So reach human level intelligence within a decade. That may be optimistic, right?

Five to ten years would be if everything goes great, all the plans that we've been making will succeed, we're not going to encounter unexpected obstacles, but that is almost certainly not going to happen. You don't like that, right?

[Speaker 2] (1:15:21 - 1:15:27)

Do you think AGI and human level intelligence you think is far, far away or unlikely?

[Speaker 1] (1:15:27 - 1:16:21)

No, I don't think it's that far away. I don't think my opinion about how far it is is very different from what you will hear from some Altman or Demis Hassabis or things like this. It's quite possibly within a decade, but it's not going to happen next year.

It's not going to happen in two years. It's going to take longer, and so you don't want to extrapolate the capabilities of LLM and say, we're just going to scale up LLM, train them with bigger computers and more data and human level intelligence is going to emerge. It's not going to work this way.

We're going to have to have those new architectures, those JEPAs, systems that learn from the real world and can plan hierarchically, can plan a sequence of actions as opposed to just producing one word after the other essentially without thinking. So system two instead of system one. LLMs are system one.

The architecture I'm describing, which I call objective-driven AI, is system two.

[Speaker 2] (1:16:22 - 1:16:56)

I'd love to come do a course at your college and learn if you'll have me as a student. I don't know if I qualify. I'll have to go back and finish high school, but I would love it.

Just to finish the LLM loop, because it's in the news and everybody's talking about LLMs. So you define a problem, you find a large data set. Most of the time goes in cleaning the data. You choose a model, you train the model, and then you execute the model.

Before that, you fine-tune the model. Before that, you fine-tune the model, yes.

[Speaker 1] (1:16:56 - 1:18:49)

What will change here? So there's still going to be a need for collecting data and filtering data to keep high-quality data and basically get rid of junk. That's actually a pretty expensive part of the whole thing.

But I think what's going to need to happen in that respect is that currently the LLMs are trained with a combination of publicly available data and licensed data, basically. But it's mostly publicly available data, publicly available text on the internet. And it's extremely biased in many ways.

A lot of it is in English. There's a significant amount of data in commonly spoken languages like Hindi, but not so much in all 22 official languages of India, and certainly not in all the 700 dialects

or whatever the number is, particularly since most of those dialects are not written, so only spoken. So what we need in the future is data sets that are more encompassing, so that the systems that are trained with it understand all the world's languages, all the world's cultures, all the value systems, everything.

And no single entity, I think, would be able to do this. Which is why I think the future of AI is, AI is going to become a kind of common infrastructure which people will use as a repository of all human knowledge. And this cannot be built by a single entity.

It's going to have to be a collaborative project with training being distributed all around the world so that you can have models that are trained on all data around the world, but you don't have to copy the data anywhere.

[Speaker 2] (1:18:49 - 1:19:16)

And a private digression. I was reviewing a data center business to invest into. A lot of people tell me that compute as a commodity will soon be sold outside of the data center and not inherently in it.

Is it a good place to focus energy and time on, like building data centers out of India? I'm taking the sovereign AI model where every country will probably fight to retain their data a bit more than they're doing currently.

[Speaker 1] (1:19:17 - 1:20:49)

Yeah, so in that kind of future, which also I alluded to with the distributed training of models, having local computing infrastructure, I think, is very important. So yes, I think that's kind of crucial. It's crucial for two reasons.

One is having local ability to train models, okay? And the second one is having very low-cost access to inference for AI systems. Because if you want AI systems to be used by, I don't know, 800 million Indians, right?

I know there are more Indians than this, but not everybody will use AI systems. But it's a lot of computing infrastructure. It's actually much bigger than the infrastructure for learning.

And there is this scenario for which there is a lot more innovation than training. Training is dominated by media at the moment. There's going to be other players, but they have a hard time competing because of the software stack, basically.

Their hardware may be really good, but the software stack is a challenge. For inference, though, there's a lot more innovation there. And that innovation is bringing down the cost.

I think the cost of inference for LLM has gone down by a factor of 100 in two years. I mean, it's

amazing, right? It's way faster than Moore's law.

And I think there is still a lot of room for improvement, and you need that because you basically need the inference for a million tokens to be a few rupees. So that's the future if you want to deploy AI assistance widely in India.

[Speaker 2] (1:20:50 - 1:21:18)

I want to use the time, Jan, because I realize we are running out, to bring it into the Indian context. Are people watching this, like I said, are entrepreneurs in play or people trying to be entrepreneurs? As an Indian 20-year-old who wants to build a business in AI, a career in AI, what do we do as we sit today?

[Speaker 1] (1:21:19 - 1:21:29)

A 20-year-old today, I would cross my finger so that when I graduate at 22, there will be good PhD programs in India.

[Speaker 2] (1:21:29 - 1:21:32)

Outside of the academic lens, I mean more…

[Speaker 1] (1:21:32 - 1:21:53)

No, no, but that's what I need to train myself to innovate. Doing a PhD or graduate studies, it trains you to invent new things and also make sure that the methodology you use prevents you from fooling yourself into thinking you're being an innovator, but you're not. So you learn this.

[Speaker 2] (1:21:54 - 1:21:57)

What if I'm an entrepreneur, a 25-year-old entrepreneur?

[Speaker 1] (1:21:57 - 1:22:42)

You still want to do a PhD if you're an entrepreneur, or at least a master's. You might be doing this by yourself, you don't have to, but it's useful because you learn more about what exists out there, what's possible, what's not possible. You get more legitimacy in hiring talented people.

There's a lot of advantages, particularly in a complex, deeply technical area like AI. You might succeed if you don't, that's not the issue, but it gives you a different perspective. But now you're doing your PhD, you're doing your startup, it might be easier to raise money if you've published a few papers where you've invented something new and you say, well, this is a new technique that really may make a difference.

[Speaker 2] (1:22:42 - 1:23:09)

What if I were to even go one step further? Let's say intelligence, I'm going to leave the AGI side of it. Let's say narrow intelligence.

Self-driving cars, robots, all of that. What should I build in? If I have to pick a subset where I can use narrow intelligence through any of the models that we spoke about, what would I start which has a capitalistic leg to it?

[Speaker 1] (1:23:09 - 1:23:50)

Okay, so today, like right now, the most likely business model that has to do with AI is taking an open source foundation model, like Lama, which is the open source system which is used everywhere now. Almost every startup uses it, even large companies. So take an open source platform, whether it's an LLM or an image feature extraction system or a segmentation system, whatever, and then fine-tune it for a particular vertical application and become an expert in that vertical application.

And which vertical should we pick? Any vertical.

[Speaker 2] (1:23:51 - 1:24:02)

But I want to know, give me top three. We interviewed Gates recently and he said, focus on building that layer around law because the legal processes arrive for disruption.

[Speaker 1] (1:24:02 - 1:26:13)

That's a good example, right? If you had to pick one or two more? I mean, in sort of B2B, there is legal, accounting, business information, right?

I want some report on the competitive situation in a particular segment of the market, you know, fintech, finance. I mean, those are obvious areas. You know, LLM's information system, they give you all the private information inside of a company so that any employee can ask any question about anything, administrative or whatever, and you get the answer.

You don't have to plow through, like, multiple internal websites and information systems. So that's certainly a good thing. And I think there is a lot of work there in basically companies that can fine-tune models for particular verticals.

And then there is other markets that are more consumer, assistance for various things. You know, for education, there's not a huge amount of money there, unless you can get contracts from the government. But education certainly is a wide application.

Probably the other big one is health. So there are a lot of companies, particularly in the developing world, that are being formed for using LLM to provide assistance, medical assistance, essentially. You know, you call up your LLM, and you say, well, I have those symptoms, should I

go to the hospital?

Or, like, this is an issue I have. And it's much easier than to get an appointment with a doctor. There's certain parts of the world where it's basically impossible to see a real doctor.

You have to travel to a city or something. So I think that would be useful. Other applications in rural areas, particularly things that are enabled by AI assistance, they can speak local languages and serve people who are not particularly comfortable with literacy, that don't write very much, don't read very much.

So interacting in your own language through speech with an AI assistant, I think, opens up a lot of applications in agriculture and, you know, all kinds of areas.

[Speaker 2] (1:26:13 - 1:26:28)

And if I were to switch the lens from an entrepreneur to an investor, what would an investor benefit from investing into AI? Would it be NVIDIA, Llama Meta, ChatGPT, OpenAI?

[Speaker 1] (1:26:29 - 1:26:43)

Okay, so I think the first order of thing is imagine what the future is going to be five years from now. And basically, that's going to be the— I suspect you would do a much better job at imagining the future than I.

[Speaker 2] (1:26:43 - 1:26:47)

Can you depict a future five years from now?

[Speaker 1] (1:26:48 - 1:27:15)

So five years from now, the world is going to be dominated by open-source platforms. For the same reason that the world of embedded devices and operating systems is dominated by Linux. The entire world runs on Linux, and it wasn't the case 20 years ago, 25 years ago.

And it's become so because open-source platforms are more portable, they're more flexible, they're more secure, they're cheaper to deploy.

[Speaker 2] (1:27:15 - 1:27:26)

I shouldn't take credit for this, but we have somebody called Kailash, who's our CTO, who's a big proponent of this. And everything we do is open-sourced. We have a fund which gives grants to open-source companies and stuff like that.

[Speaker 1] (1:27:26 - 1:27:59)

Right, okay, so the world is going to be open-source. We're going to have open-source AI

platforms. In a few years, they'll probably be trained in a distributed fashion, so they're not going to be completely controlled by a single company.

The proprietary engines, I think, are not going to be nearly as important as they are today, because the open-source platforms are catching up in terms of performance. And then what we know is that a fine-tuned open-source engine like Lama always works better than a non-fine-tuned generic top-performing model.

[Speaker 2] (1:28:00 - 1:28:07)

But if everything is open-sourced, it'll also be democratized for an investor to invest into, then what is the differentiation?

[Speaker 1] (1:28:07 - 1:28:33)

Well, it enables the ecosystem. If you are a startup, you're much better off using an open-source engine and fine-tuning it for a vertical application than you are using an API, because you can build a tailored product for your customers in a much better way. So that's the first thing.

Second thing is, if you really want this technology to be democratized and used by everyone, eventually using smart glasses and stuff, but at first just smartphones.

[Speaker 2] (1:28:33 - 1:28:41)

Do you think the form will change soon? The form of how you interact with technology will move from smartphones to different kind of devices soon?

[Speaker 1] (1:28:41 - 1:29:09)

Smart glasses, yeah. I mean, yeah, there's almost no question. You're using one.

So I don't have them right now, although they are in my bag right here. I use them all the time, yeah. I find them really useful for all kinds of stuff, even if you don't use AI for it, just taking pictures or listening to music or whatever, but then you have the AI assistant and I could be sitting in a restaurant with a menu in a foreign script in a foreign language and they could be translating it for me.

[Speaker 2] (1:29:10 - 1:29:21)

So what happens to intelligence in society with all of this changing? What becomes, forget computers and AI for a second, for humans, what is intelligence in that world?

[Speaker 1] (1:29:22 - 1:30:14)

So people's intelligence will be moving to a different set of tasks than the one we are trying to do

today. Because a lot of what we're trying to do today will be done by AI systems. And so we will focus on other tasks.

So things like not doing things, but deciding what to do or figuring out what to do, okay? Those are two different things. Like, think about the difference between a low-level employee in a company that is told what to do and just does it, and then a high-level manager in the company that has to figure out strategy and think about what to do, and then tell people what to do.

We're all going to be a boss. We're all going to be like those high-level managers. We're going to tell our AI systems what to do, but we're not going to have to do it ourselves, necessarily, okay?

[Speaker 2] (1:30:14 - 1:30:24)

But we need lesser people to tell something more efficient than us what to do than we need today for them to actually do the task, right? Yeah. So what happens to everyone else?

[Speaker 1] (1:30:26 - 1:31:52)

Well, I think everyone is going to be in that situation. It's going to have access to AI systems and be able to delegate a lot of tasks, you know, mostly in the virtual world, but eventually in the real world. We're going to have, at some point, domestic robots and self-driving cars and things like this once we figure out how to get the system to learn how the real world works from video.

So the type of tasks on which we are going to be able to concentrate ourselves are going to be more abstract. The same way, you know, nobody needs to do, like, super-fast mental arithmetics anymore. We have calculators.

Or, you know, solve integrals and differential equations. We have to learn the basics before we do this, but we can use computer tools to do this, right? So it's going to lift the abstraction level at which we can place ourselves and basically enable us to be more creative, be more productive, okay?

And there are a lot of things that you and I have learned to do that our descendants would not have to learn to do because they would be taken care of by machines. Go to school. Like, go to school?

No, no, we'll still go to school. We'll have to educate ourselves. There's still going to be the competition between humans to kind of do something better than the others or something different, more creative.

[Speaker 2] (1:31:52 - 1:31:54)

Always, right? Innately, we don't want to compete with our peer groups.

[Speaker 1] (1:31:54 - 1:32:07)

Yeah, yeah. So we're not going to run out of jobs. Economists that I talk to tell me we're not going to run out of jobs because we're not going to run out of problems.

But we're going to find better solutions to problems with the help of AI.

[Speaker 2] (1:32:07 - 1:32:18)

Maybe we can end today, Jan, trying to define what is intelligence really. I had written down intelligence as a collection of information and the ability to absorb new skills.

[Speaker 1] (1:32:19 - 1:33:15)

It's a collection of skills and an ability to learn new skills really quickly or an ability to solve problems without learning. This is called zero shot in the AI business. You're faced with a new problem and you can think about it for a while and you may never have faced a similar problem before, but you can solve it by just thinking and using your mental model of the situation.

That's called zero shot. You're not learning a new skill. You're just solving a problem from scratch.

So the combination of those three things, having already a number of skills that you know, experience with solving problems and accomplishing tasks, being able to learn new tasks really quickly with a few trials, and then the next step is being able to solve new problems, zero shot, without having to learn anything new. That's the combination of those three things really is intelligence.

[Speaker 2] (1:33:16 - 1:33:31)

Thank you, Ian, so much for doing this. I'm going to try and figure out how I can do a course under you wherever you're teaching. Maybe you can recommend me to the college to give me a seat so I can attend some lectures.

You can do better.

[Speaker 1] (1:33:32 - 1:33:46)

The 2021 edition of my deep learning course is fully available on the internet for free. It's all on YouTube, all the problems, all the homework and everything. I feel like I'm going back to old school.

[Speaker 2] (1:33:46 - 1:33:57)

I feel like being in front of you and learning, first person has an innate value of its own. So I'll try and do this and that. Wonderful.

Thank you so much, Ian, for doing this.

[Speaker 1] (1:33:58 - 1:33:58)

My pleasure.

[Speaker 2] (1:34:04 - 1:34:08)

Thank you. That was fun. That was fun, yeah.

You didn't get bored?

[Speaker 1] (1:34:10 - 1:34:17)

No. You ask a professor to speak, like, you know. That's the job.

[Speaker 2] (1:34:19 - 1:34:24)

But I guess, like, when you're talking to people who know so much less than you, it can't be fun all the time.

[Speaker 1] (1:34:24 - 1:34:35)

It's an art. I mean, I don't claim to be particularly good at it, but I try hard. So, you know, trying to kind of simplify concepts and stuff like that.

[Speaker 2] (1:34:36 - 1:34:45)

But I think we needed it because so many Indians are speaking about AI, but so few of us actually understand what went behind where we are today.

[Speaker 1] (1:34:45 - 1:34:55)

Oh, that's true across the world. It's not just India. In fact, I think it's kind of the opposite in India.

It's, like, way more people who are kind of educating themselves, like, particularly among the young people.

[Speaker 2] (1:34:56 - 1:35:08)

So we wanted to focus today on that, like, just to get to telling our people, a lot of young people watch this, young, bright people, how we got to be where we are today. The most questions around that.

[Speaker 1] (1:35:08 - 1:35:55)

Yeah, I think it's important because it helps convince people that you can do it regardless of

what... Like, you know, I went to... I studied engineering in France, but I didn't go to, like, one of the top, you know, equivalent Ivy League or anything.

I went to, like, a regular school, right? And I didn't do my PhD with a famous person and, you know, all that stuff. When I was in France, I was writing papers in French that were terrible, nobody read, but, you know, kind of managed to do something.

And people are sometimes telling me, like, you're... You know, you helped convince me that I could do something impactful even though I didn't go to Harvard or MIT or Stanford. Thank you.